GROUP11

BSIT-S-4A



MOTIVANCE: A Mobile Computer Vision-Based Physical Rehabilitation Support System with Exercise Guidance, Range of Motion Assessment, and Patient Progress Monitoring for Mild Stroke Patients

SUMMARY

Motivance is a mobile computer vision-based physical rehabilitation support system that uses a smartphone camera to analyze patient movements without wearable markers or specialized equipment.

It detects body landmarks, estimates joint angles and range of motion (ROM), counts repetitions, and evaluates exercises based on therapist-defined parameters and ROM thresholds.

The system also provides real-time exercise guidance and records ROM, repetitions, and session data to track rehabilitation progress.

Physical therapists can manage patients, assign exercises, set movement targets, and review performance, while Motivance serves as a support tool rather than a replacement for professional healthcare assessment.

GENERAL OBJECTIVE

To develop **Motivance**, a mobile computer vision-based physical rehabilitation support system that assists physical therapists in assessing joint movement and range of motion, providing real-time exercise guidance, managing patient rehabilitation activities, and tracking rehabilitation progress based on professionally defined movement thresholds.

SPECIFIC OBJECTIVES

- 1. To develop a markerless computer vision module that detects and tracks relevant body landmarks through a smartphone camera during selected physical therapy exercises.

- 2. To develop a movement assessment module that estimates joint angles, range of motion, repetitions, movement duration, and other measurable movement parameters during supported exercises.

- 3. To develop a rehabilitation management module that allows physical therapists to manage patient information, assign exercises, and establish individualized movement and range-of-motion thresholds.

- 4. To develop a real-time exercise guidance module that compares the patient's detected movement with predefined exercise parameters and provides visual feedback regarding range of motion, repetitions, and supported movement-form conditions.

SPECIFIC OBJECTIVES

- 5. To develop a remote exercise support module that allows patients to perform therapist-assigned exercises using their smartphones and records their exercise performance for subsequent professional review.

- 6. To develop a progress-tracking module that records rehabilitation sessions and presents changes in range of motion, repetitions, and exercise performance across sessions.

- 7. To evaluate Motivance in terms of movement-measurement accuracy, usability, performance, reliability, and its effectiveness in supporting physical therapy management.

FUNCTIONAL REQUIREMENTS

1. User and Patient Management

- The system shall provide separate accounts for patients, physical therapists, and administrators.
- The system shall provide user registration and login.
- The system shall implement role-based access.
- The physical therapist shall be able to register and manage patient profiles.
- The physical therapist shall be able to view the rehabilitation history of assigned patients.
- The system shall maintain patient exercise and session records.

2. Camera and Movement Setup

- The system shall provide instructions for proper smartphone positioning.
- The system shall determine whether the patient is positioned within the camera's detectable area.
- The system shall detect relevant body landmarks using the smartphone camera.
- The system shall identify the joints required for the selected exercise.
- The system shall perform camera positioning or calibration before movement assessment.

FUNCTIONAL REQUIREMENTS

3. Movement and ROM Assessment

- The system shall detect and track relevant body landmarks during supported exercises.
- The system shall calculate the estimated joint angle.
- The system shall determine the patient's range of motion.
- The system shall detect and count exercise repetitions.
- The system shall record exercise duration.
- The system shall record measurable movement performance during exercise sessions.

4. Exercise Assignment and Threshold Management

- The physical therapist shall be able to assign exercises to patients.
- The physical therapist shall be able to establish the required ROM or movement threshold.
- The physical therapist shall be able to establish the required number of repetitions.
- The physical therapist shall be able to configure parameters for supported exercises.
- The system shall associate exercises and movement thresholds with the appropriate patient.

FUNCTIONAL REQUIREMENTS

5. Real-Time Exercise Guidance

- The system shall display instructions for the assigned exercise.
- The system shall display the required ROM or movement threshold.
- The system shall display detected movement information in real time.
- The system shall indicate whether the patient has reached the required movement range.
- The system shall detect incomplete repetitions.
- The system shall provide feedback when supported exercise-form conditions are not satisfied.
- The system shall provide visual feedback during exercise execution.

6. Remote Exercise Support

- The patient shall be able to access exercises assigned by the physical therapist.
- The patient shall be able to perform assigned exercises using a smartphone camera.
- The system shall record movement data during remote exercise sessions.
- The system shall save completed session results.

FUNCTIONAL REQUIREMENTS

7. Rehabilitation Progress Tracking

- The system shall store completed exercise sessions.
- The system shall record ROM measurements.
- The system shall record completed repetitions.
- The system shall compare exercise performance across sessions.
- The system shall display ROM progression through charts or visualizations.
- The system shall provide historical rehabilitation performance for therapist review.

8. Physical Therapist Dashboard

- The physical therapist shall be able to view assigned patients.
- The physical therapist shall be able to assign exercises.
- The physical therapist shall be able to establish movement thresholds.
- The physical therapist shall be able to review patient exercise results.
- The physical therapist shall be able to view patient progress.

NON-FUNCTIONAL REQUIREMENTS

Accuracy

- The system should provide consistent joint-angle and ROM measurements under defined testing conditions.
- The system should identify and report limitations affecting movement measurements.
- The movement measurements should be evaluated against an appropriate reference measurement.
- The system should distinguish between successfully detected movements and situations where reliable movement detection cannot be performed.

Performance

- The system should provide real-time movement feedback with minimal delay.
- Normal application operations should respond within approximately three seconds.
- Computer-vision processing should operate efficiently on supported smartphones.
- The system should maintain stable performance during supported exercise sessions.

NON-FUNCTIONAL REQUIREMENTS

Usability

- The interface should be simple and understandable for elderly users and individuals with limited technical experience.
- Buttons and text should be sufficiently large and readable.
- Exercise instructions should be clear and easy to follow.
- Real-time feedback should be presented in an easily understandable manner.
- The system should support Filipino and English where applicable.
- The system should minimize the number of steps required to begin an assigned exercise.

Reliability

- The system should prevent loss of rehabilitation records.
- The system should handle temporary camera interruptions.
- The system should handle temporary network interruptions where applicable.
- The system should safely recover from application errors.
- Completed exercise-session data should be saved consistently.

NON-FUNCTIONAL REQUIREMENTS

Security and Privacy

- The system should use secure authentication.
- Patient information should be protected through role-based access.
- Communication involving patient information should be encrypted.
- The system should obtain patient consent for camera and data collection.
- The system should minimize unnecessary storage of raw video data when movement metrics are sufficient.

Compatibility

- The system should primarily support Android smartphones.
- The system should support commonly available smartphone cameras.
- The system should accommodate supported smartphone screen sizes.
- The system should function under the defined camera positioning and environmental conditions.

NON-FUNCTIONAL REQUIREMENTS

Maintainability

- The computer vision, mobile application, rehabilitation management, database, and backend components should be modular.
- New supported exercises should be addable without redesigning the entire system.
- Additional joints should be extendable in future versions.
- Changes to exercise parameters should not require modification of the core computer-vision module.

SIMULATION



https://stitch.withgoogle.com/preview/18125759532129365195?node-id=9ad652bdb3254c76bbae556bb2789ca2

REASON FOR CHOOSING THIS PROJECT

We chose **Motivance** to address challenges faced by physical therapists in patient management, documentation, and manual range-of-motion assessment.

The system uses a smartphone camera and computer vision to estimate joint movement, provide real-time exercise guidance based on therapist-defined thresholds, and record rehabilitation performance.

It also supports remote exercise sessions and progress tracking, particularly for elderly patients and individuals with mobility limitations. **Motivance** is designed to support—not replace—physical therapists by reducing manual tasks and providing accessible rehabilitation information.

CLIENT VISIT (Taguig Disability Resource and Development Center

September 4, 2026



October 15, 2026



PRE-EVALUATION

GALAWGO

PRE-EVALUATION FORM

Mobile Computer Vision-Based Physical Rehabilitation Support System

Purpose: To determine whether the proposed features of GalawGo meet the needs of patients and physical therapists before system development or implementation.

Instructions: Place a check mark (✓) in only one box for each statement.

Name (Optional): STEPHANIE JOANNA MARCELINO Date: 10/15/26

Role: ☐ Patient ☐ Physical Therapist ☐ Other: _____

No.	FUNCTIONAL REQUIREMENT	Yes	No	Maybe
A. USER AND PATIENT MANAGEMENT				
1	The system should allow users to register and log in.	☑	☐	☐
2	The system should provide separate access for patients, physical therapists, and administrators.	☑	☐	☐
3	Physical therapists should be able to manage patient profiles and view their rehabilitation history.	☑	☐	☐
B. CAMERA AND MOVEMENT SETUP				
4	The system should provide instructions for properly positioning the smartphone camera.	☑	☐	☐
5	The system should detect whether the patient is properly positioned within the camera's view.	☑	☐	☐
6	The system should detect and track body movements using the smartphone camera without wearable markers.	☑	☐	☐
C. MOVEMENT AND ROM ASSESSMENT				
7	The system should measure estimated joint angles and range of motion (ROM).	☑	☐	☐
8	The system should automatically count exercise repetitions.	☑	☐	☐
9	The system should record exercise duration and movement performance.	☑	☐	☐
D. EXERCISE ASSIGNMENT AND GUIDANCE				
10	Physical therapists should be able to assign exercises to their patients.	☑	☐	☐
11	Physical therapists should be able to set required ROM targets and repetition counts.	☑	☐	☐
12	The system should display instructions while the patient performs an exercise.	☑	☐	☐
13	The system should provide real-time visual feedback about the patient's movements.	☑	☐	☐
14	The system should identify incomplete repetitions or movements that do not meet supported exercise requirements.	☑	☐	☐
E. REMOTE EXERCISE SUPPORT				
15	Patients should be able to access exercises assigned by their physical therapist using a smartphone.	☑	☐	☐
16	The system should record and save the results of completed exercise sessions.	☑	☐	☐
17	Physical therapists should be able to review patients' remotely recorded exercise performance.	☑	☐	☐
F. REHABILITATION PROGRESS TRACKING				
18	The system should save previous exercise sessions and ROM measurements.	☑	☐	☐
19	The system should compare exercise performance across different sessions.	☑	☐	☐
20	The system should display rehabilitation progress through charts or visualizations.	☑	☐	☐
G. PHYSICAL THERAPIST DASHBOARD				
21	Physical therapists should have a dashboard to view their assigned patients.	☑	☐	☐
22	Physical therapists should be able to review patient exercise results and progress in one place.	☑	☐	☐
23	Physical therapists should be able to review the rehabilitation history of their patients.	☑	☐	☐

PART II. OVERALL ASSESSMENT

No.	STATEMENT	Yes	No	Maybe
24	GalawGo could help make rehabilitation exercises easier to monitor.	☑	☐	☐
25	GalawGo could help reduce the manual work involved in recording exercise results.	☑	☐	☐
26	GalawGo could help patients monitor their rehabilitation progress at home.	☑	☐	☐
27	GalawGo could be useful as a support tool for physical therapists.	☑	☐	☐
28	I think the proposed functionalities of GalawGo are useful and worth developing.	☑	☐	☐

PART III. ADDITIONAL COMMENTS (OPTIONAL)

1. What other features would you like GalawGo to have?
   Hopefully pt could also provide their feedback regarding exercises
2. What concerns or difficulties do you expect when using GalawGo?
   Age, hopefully the app is easy to function by the pt.

Thank you for your participation!

This form gathers initial opinions about proposed features; it does not measure actual system accuracy or performance.

NDA

NON-DISCLOSURE AGREEMENT

For Thesis Needs Assessment Interview

This Non-Disclosure Agreement ("Agreement") is entered into on SEPT. 4, 2026, by and between:

RESEARCHERS:
Group 11 – BSIT-S-4A
Technological University of the Philippines – Taguig

- Ramon Francisco Jr.
- Ronald Ajusan
- Aeron Jhon Canta
- Decibelle Reign Tanora

and

CLIENT/RESPONDENT:
Taguig Physical Medicine and Rehabilitation Unit
Represented by: FRANCESCA FAJARDO / MARCELINO, STEPHANIE PAULA
Position: PHYSICAL THERAPIST

Together, the Researchers and the Client/Respondent shall be referred to as the "Parties."

1. Purpose of the Agreement

This Agreement is intended to protect confidential information that may be shared during the needs assessment interview conducted in relation to the thesis project entitled:

"iMove: A Mobile Computer Vision-Based Rehabilitation Support System for Joint Movement Assessment Using Wearable Visual Markers."

The information obtained will be used solely to understand the rehabilitation needs, current practices, challenges, and requirements of the intended client and to guide the researchers in defining and developing the proposed system.

2. Confidential Information

For purposes of this Agreement, "Confidential Information" refers to any non-public information shared during the interview or related discussions, including but not limited to: